Project Report: Emotion-Driven Playlist Generator Using EEG Signals

1. Introduction

This project aims to build a real-time music recommendation system that uses brain signals (EEG) to detect emotional states and play corresponding Spotify playlists.

2. Problem Statement

Traditional music recommendation systems rely on listening history and user preferences. They fail to capture real-time emotional changes. This project addresses that gap by using EEG signals to determine the user's current emotional state.

3. Objectives

- Simulate or use real EEG signals.
- Train a model to classify emotional states from EEG.
- Use Spotify API to play playlists based on detected emotion.
- Build a Flask web app with TailwindCSS interface.

4. Dataset Creation

4.1 EEG Data Generation (Simulated)

Generate a balanced synthetic EEG dataset for 6 emotions to train an emotion classification model.

Emotions Included:

- 😊 **happy**
- 😢 **sad**

- 😌 **relaxed**
- 😠 **angry**
- 😨 **fearful**
- 😲 **surprised**

Data Details:

- Samples per emotion: 300
- Total samples: 6 × 300 = 1800
- Features: 8 EEG channels (ch1 to ch8)
- Label column: label (emotion name)
- Value range: Each channel has values in the range [0, 3], simulating real EEG amplitudes.

Output:

Saves a CSV file named eeg_data.csv with 1800 rows and 9 columns.

	A	B	C	D	E	F	G	H	I
1	ch1	ch2	ch3	ch4	ch5	ch6	ch7	ch8	label
2	2.526	2.566	2.91	2.37	2.778	2.635	2.769	2.55	happy
3	0.706	0.769	0.769	0.404	0.388	0.576	0.435	0.69	sad
4	1.999	2.136	2.291	1.798	2.5	2.054	2.404	2.101	angry
5	2.579	2.807	2.878	2.471	2.87	2.675	2.713	2.367	happy
6	1.646	1.403	1.509	1.361	1.648	1.588	1.302	1.557	relaxed
7	1.378	1.481	1.668	1.414	1.681	1.59	1.357	1.603	relaxed
8	2.186	2.371	2.288	2.057	2.466	2.411	2.409	2.196	surprised
9	2.126	2.393	2.276	2.11	2.381	2.261	2.378	2.106	surprised
10	1.365	1.54	1.47	1.32	1.708	1.575	1.487	1.586	relaxed
11	1.49	1.547	1.606	1.273	1.687	1.5	1.494	1.464	relaxed
12	1.412	1.362	1.54	1.32	1.824	1.704	1.387	1.484	relaxed

5. Model Training

Train a neural network (EEGNet) to classify **6 emotional states** using simulated EEG data.

📊 Dataset

- **Source:** eeg_data.csv
- **Features:** 8 EEG channels (ch1 to ch8)
- **Labels:** Encoded from text (happy, sad, relaxed, angry, fearful, surprised) to integers
- **Split:** 80% train, 20% test
- **Samples used:** 1,200 (200 per emotion)

🧠 Model: EEGNet

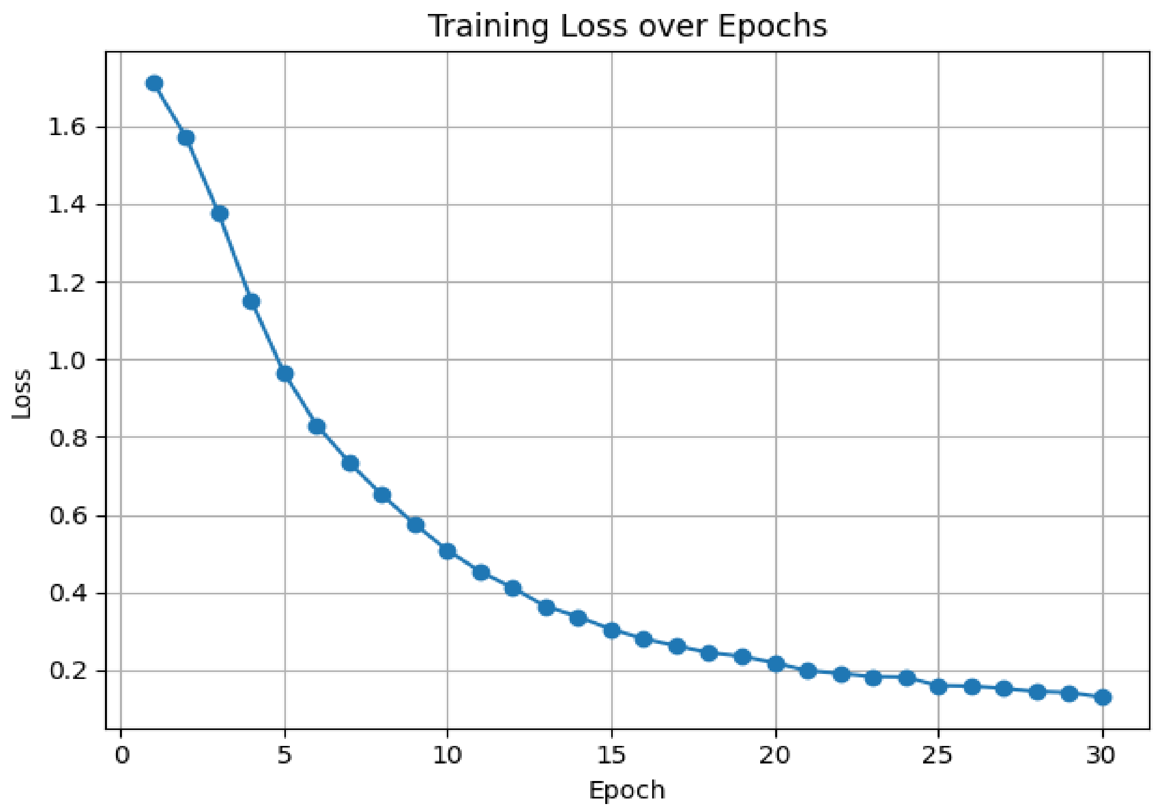
Layer	Type	Size
Input	Linear(8, 64)	8 EEG features
Hidden	ReLU	Non-linearity
Hidden	Linear(64, 32)	
Hidden	ReLU	
Output	Linear(32, 6)	6 emotion classes

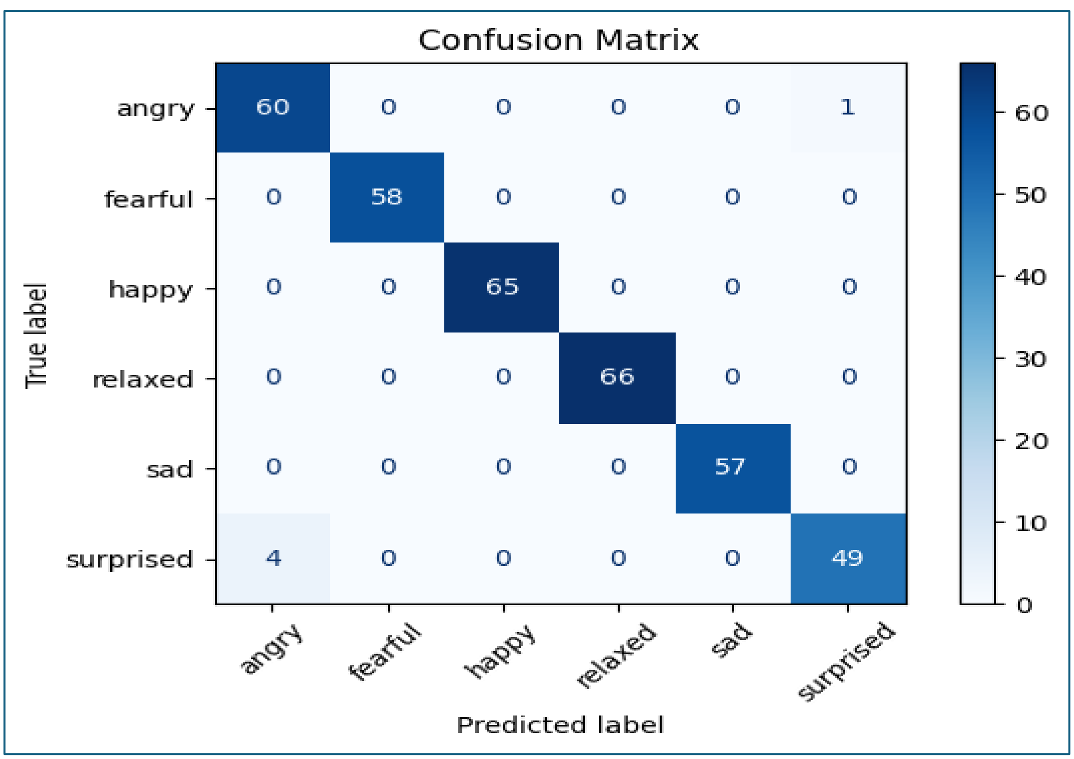
Training

- **Loss Function:** CrossEntropyLoss
- **Optimizer:** Adam
- **Epochs:** 30
- **Batch Size:** 32

Output

- Trained weights saved as: **emotion_model.pt**





6. Web App (Flask + Tailwind)

Build a web interface that simulates EEG input, predicts emotional state using a trained PyTorch model, and opens a relevant Spotify playlist.

🔧 Technologies Used

- **Flask**: For web backend and routing
- **PyTorch**: For loading the trained emotion_model.pt
- **Webbrowser module**: To auto-open the playlist in the user's browser
- **HTML (Jinja)**: Rendered using index.html

Model

- **Model Used**: EEGNet with structure:

Linear(8 → 64) → ReLU → Linear(64 → 32) → ReLU → Linear(32 → 6)

- **Input**: 8 simulated EEG channels
- **Output**: One of 6 emotions

Each emotion maps to:

- An emoji
- A Spotify playlist cover image
- A Spotify playlist URL

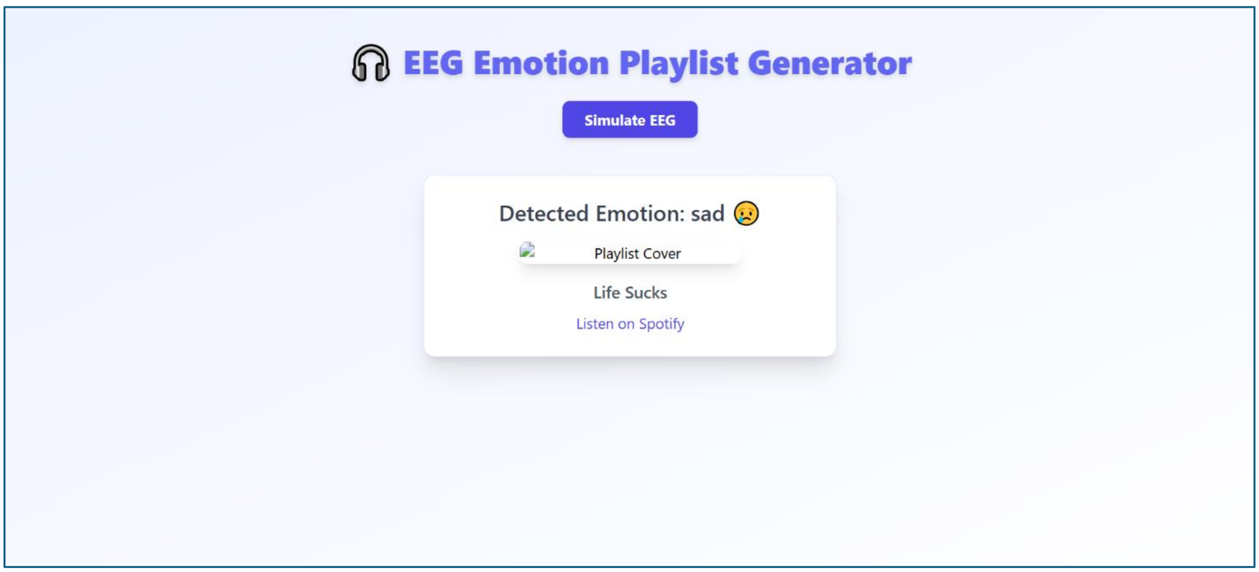
⚙️ Routes

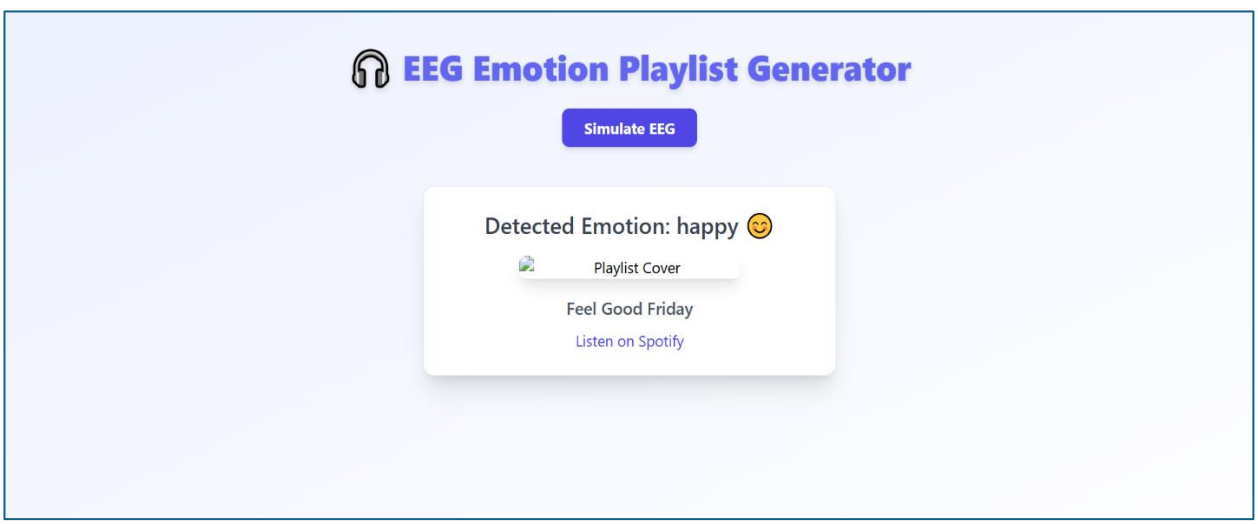
- / → Renders the main page
- /predict →
    - Simulates EEG signal
    - Predicts emotion using the model

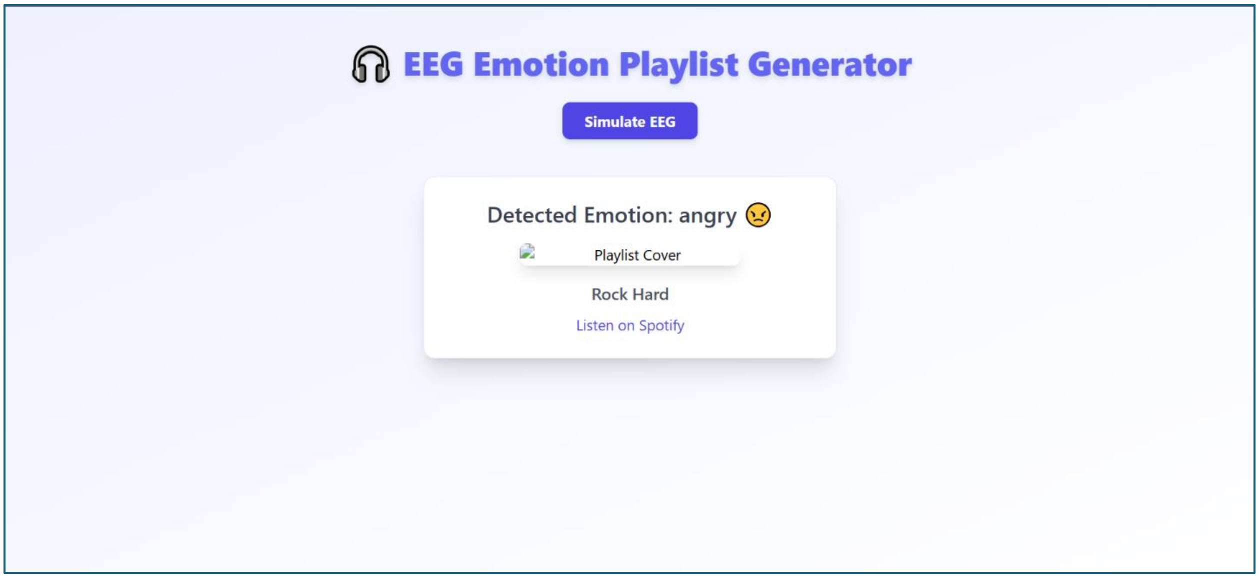
- Opens playlist URL in browser
- Displays emoji and playlist info on page

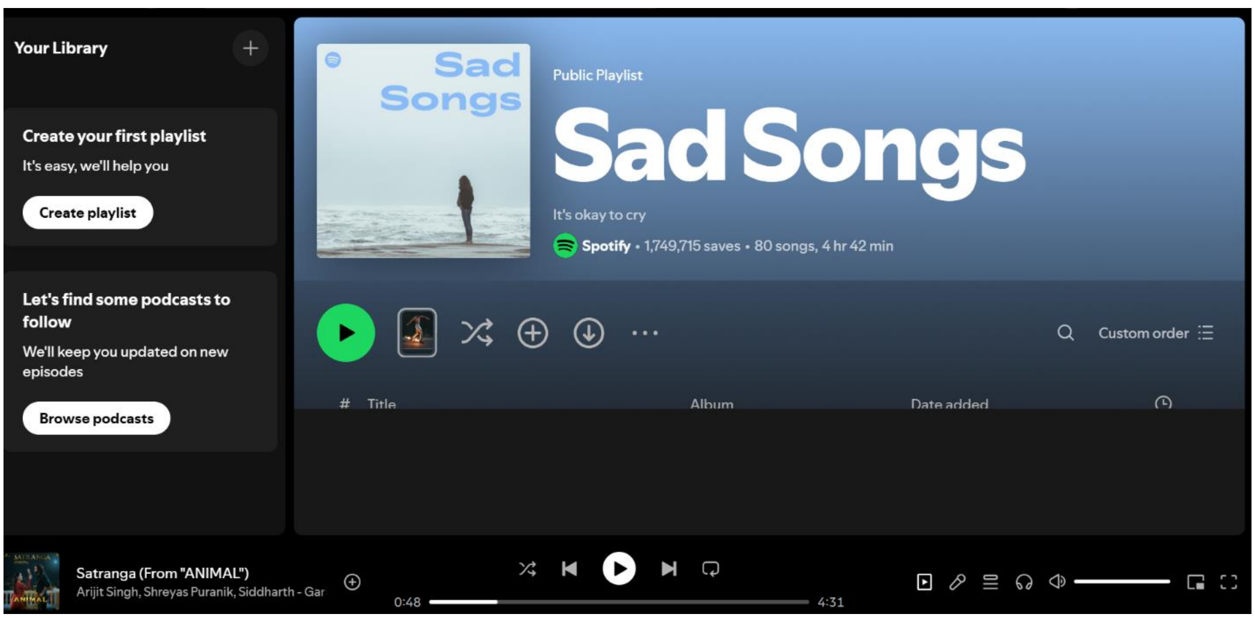
💡 Output

- GUI displays:
  - Predicted emotion
  - Emoji
  - Playlist cover and name
- Auto-opens the matched Spotify playlist in browser






EEG Emotion Playlist Generator
Simulate EEG
Detected Emotion: angry
Playlist Cover
Rock Hard
Listen on Spotify


Your Library
Create your first playlist
It's easy, we'll help you
Create playlist
Let's find some podcasts to follow
We'll keep you updated on new episodes
Browse podcasts
Sad Songs
Public Playlist
Sad Songs
It's okay to cry
Spotify • 1,749,715 saves • 80 songs, 4 hr 42 min
Custom order
Title
Album
Date added
Satranga (From "ANIMAL")
Arijit Singh, Shreyas Puranik, Siddharth - Gar
0:48
4:31